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An intelligent energy efficient cooperative MIMO-AF multi-hop and relay based communications for Unmanned Aerial Vehicular networks

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) are recently used for both civilian and military applications in worldwide. Energy Efficiency (EE) is an exceptional design approach for modern communication based systems. New advance technology is needed in order to support UAV applications with reduced energy usage. In this paper, an Energy Efficiency with Hybrid Fuzzy Firefly Algorithm (EE-HFFA) method is introduced for Multiple-Input–Multiple-Output (MIMO) Amplify-And-Forward (AF) systems in which Partial Channel State Information (PCSI) estimation is existing at the relays because of the high speed mobility. A new EE-HFFA algorithm is presented in this research, by means of merging the benefits of the Firefly Algorithm (FA) as well as Differential Evolution (DE). For increasing information sharing both the techniques are implemented in parallel and as a result improve searching efficiency. The outcomes of these two techniques are based upon the fuzzy membership function. For approximation of PCSI for the source node as well as relay nodes, Branch Convolutional Neural Network (B-CNN) classifier is presented to raise the capability of cooperative MIMO-AF systems. EE-optimal source and relay precoding matrices are cooperatively enhanced by means of EE-HFFA. Simulation out comes illustrate that the presented EE-HFFA as well as B-CNN classifier could enhance the EE of MIMO-AF systems with PSCI while matched up with direct/relay link merely precoding optimization.

1. Introduction

The need for energy utilization in Information and Communication Technology (ICT) is rising with the high demand for services and pervasive access. All over the world billions of subscribers depend upon mobile devices for their daily lives that requires enhanced energy utilization. Energy Efficiency (EE) is progressively attained several research interest as it impacts the sustainability of communication systems for designing the future systems [1,2]. Likewise, in the last decade, relay-based cooperative communication has realized more interests because of its efficiency in improving Spectral [3], in addition to reduce the price of network deployment [4].

At present, it was illustrated in [5] that it could be useful for conserving energy in cellular systems. Relay based cooperative communication is an indispensable component in 4G [6], and a component of 5G; such as, machine-to-machine communications. To the degree that relaying methods are taken, Amplify-and Forward (AF) is one among the simplest as well as practical techniques for developing cooperative Multi-Input Multi-Output communication. MIMO is used for or SE maximizing or error rate minimizing pre-coding/resource allocation techniques [7–9]. A novel pre-coding/resource allocation designs were presented with the rising significance of EE in communication, on the other hand, this time for maximizing the EE [5,10].

The benefits of MIMO systems are enhanced not only when information about the channel, that is, channel state information (CSI), is known at the receiver but also at the transmitter [5]. When there is channel state information at the transmitter (CSIT), a processing technique, known as precoding, exploits the information about the channel by operating on the signal before transmission to better suit it for channel conditions. The transmitter cannot directly access its own CSI, thus CSIT can only be acquired indirectly. In practice, the quality of CSI obtained is often imperfect and degraded due to quantization errors, feedback delay, and channel estimation errors [10]. The imperfect CSI (ICSI) causes performance degradation and since these errors are inevitable, it is necessary to formulate robust designs to combat their effects. There have been many works on AF MIMO relaying systems are presented with certain drawbacks which are overcome by this proposed work are discussed below.

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The proposed works illustrate that EE-optimal pre-coding and allocation of resources is followed for the two-hop scenario, in which a source node transfers data to a particular node which is the destination (DN) through a relay node (RN) on the other hand deprived of considering direct transmission (that is to say SN to DN transmission is abandoned). These two techniques take up that complete Partial channel state information (PCSI), that is to say transmit as well as receive PCSI, is existing at the relay and transmit PCSI is available at the SN. Commonly, the channel state information at the transmitter, even partial information, is used to rise the channel capacity. This work was inspired by a natural insight that there is a tradeoff amid the development in channel capacity as well as the degree of completeness of the PCSI existing at the transmitter. This work inspects a new design of MIMO-AF Relay Networks under Partial Channel State Information (PCSI) by taking Channel Estimation (CE) errors as well as feedback delay errors, deprived of considering the straight link from source to destination. The complete overview of the MIMO-AF relay is provided in this part. The review work of this paper is provided in Section 2. The research technique of the MIMO-AF relay is provided in Section 3. The performance assessment of the research technique is provided in Section 4. The conclusion and future work of this paper is provided in Section 5.

2. Literature review

Hua (2010) in [11], relay classification is carried out as well as BF methods are provided for numerous mixtures of regenerative versus non-regenerative, full-duplex versus half-duplex, and one-way versus two-way. The researcher converses the construction of transmit–receive BF matrices, self-interference for full-duplex relays, amplification matrices for non-regenerative relays, in addition to consecutive interference cancellation for the two-way case. Here, descriptions as well as classifications for numerous network topologies as well as communication techniques are provided. Specially, articles that examine BF techniques for single as well as multiuser communications are provided and their key contributions are emphasized. As well, a classification dependent upon numerous relaying topologies for instance single and multiple relaying as well as opportunistic relay selection is presented as well as the equivalent BF techniques are conversed. Since the included works take channel models as well as fading distributions, which are inspected thoroughly in MIMO relaying, there is no channel modeling literature review in this survey is provided on BF techniques in addition to their numerous implementations.

Lee et al. (2010) [12] discussed about the multiple SD pairs is taken into account that communicate via MIMO AF relays as well as take on the two-way technique. The nodes are presumed to be full-duplex as well as perfect CSI is existing to all of them. Two diverse cases are provided in this perspective. The primary one is called relay channel that encompasses three users that need to unicast independent messages for two dissimilar users via the relay. The attainable Degrees of Freedom (DoF) are taken out for this setting while the relay carries out Analog Network Coding (ANC) or Physical Layer Network Coding (PLNC) and are confirmed to be equivalent to when the entire nodes contain antennas. Second one takes multiple two-way relay links work simultaneously, that specifying this case as two-way relay channels. The DoF is equivalent to 8, by means of providing the users with antennas as well as the relay with antennas. The fore most goal of this work is the examination of the DoF in two-way relay networks.

Joung and Sayed (2010) [13] illustrates the topology in which 2K MIMO users communicate in pairs via a MIMO-AF relay. The optimization problem is expressed for ZF as well as MMSE condition considering a power restraint at the relay as well as predetermined transmit–receive BF vectors that are got from CSI. As well, four BF approaches are proposed that handle diverse CSI conditions; initially, eigen-BF that needs exact CSI knowledge at the users with the intention of producing the eigen-BF vector; next, antenna selection that is carried out with partial CSI as merely the CSI of the chosen antenna is fed back to the relay; after that, random BF that does not presume CSI knowledge on the other hand requires synchronization info amongst the users as well as the relay; lastly, equal gain BF that as well does not require any CSI and no additional info at the relay. One more area, which the authors examine, is the power control for ZF systems. Two diverse cases are analyzed beginning with local power control, which dispenses the multiuser power to the users in keeping with their channel conditions and at that point global power control for the greater SNR regime divides network power to the users as well as the relay with the aim of system SNR maximization.

Chalise et al. (2012) discussed about the Partial Relay Selection (PRS) is applied, in which suboptimal relay selection is provided, since the only CSI taken in the selection algorithm is known as the quality of the SR links [14]. Also, at the source and the destination, transmit as well as receive BF is employed, when for the chosen AF MIMO relay the linear pre-coder is enhanced. Moreover, the outage probability of the technique is provided in closed form as well as for the asymptotically greater Signal Noise Ratio (SNR) the diversity gain is taken out. Consequently, PRS and OR attain the similar diversity gain.

Je et al. (2009) [15] explained a joint pre-coding technique is presented for the Base Station (BS) as well as Relay Station (RS) to rise the ergodic capability of downlink non-regenerative MIMO relay systems. Unlike from existing work, it presumes just the mean and the covariance matrix are well-known for the Channel State Information (CSI) of the RS-Mobile Station (MS) link. As the RS-MS channel is partially known, the perfect solution of this problem is hard to be identified, it use pre-coding matrices, which increase the upper bound on the ergodic capacity. Dependent upon this method, the research technique utilizes an iterative technique for the purpose of identifying BS as well as RS pre-coding matrices concurrently.

Kanakis and Rapajic (2007) illustrate the consequence of sectional channel knowledge at the receiver that produces an error in estimation at the receiver as well as the spatial correlation on MIMO Relay wireless communication systems [16]. The broadcaster contains exact insights of statistical parameters of the channel for instance covariance or mean. It presumes a Zero Mean Circular Symmetric Complex Gaussian (ZMCSCG) channel elements as well as ZMCSCG noise elements. It inspects spatial correlation at the transmitter; it presumes that the components in the rows of the channel matrix of the first hop are correlated. Partial CSI at the receiver produces bugs at the approximation of the channel at the receiver that brings about the channel capacity to worsen and to saturate at some levels of SNR.

Héliot and Tafazolli (2017) Introduced an advance method to increase the EE of non regenerative cooperative communication systems to enhance source as well as relay pre-coders during relay and direct links are taken into consideration [17]. It is confirmed that the equivalent optimization problem [18–20] is however severely pseudo-convex, that is to say containing a single solution while relay pre-coding matrix is determined and its Lagrangian is low as well as upper bounded using pseudo-convex function while source pre-coding matrix is also determined. Derive EE-optimal sources as well as relay pre-coding matrices, which could enhance via alternating optimization.

Optimization-driven methods employ suboptimal search algorithms to find the best subset. Padmanabhan et al. [21] considered the problem of receive antenna selection using the known temporal correlation of the channel symbols embedded in the data packets; the model was stated as a problem of minimizing the average packet error rate and converted to a partially observable Markov decision process framework which was solved by heuristic searching schemes. Yan et al. [22] modeled the trade-off between feedback overhead and secrecy performance by maximizing SNR of the transmitter–receiver channel. Hence the optimization approaches provides better results in which to overcome the disadvantages of the existing approaches. So, in this work, a new MIMO-AF system under Partial Channel State Information (PCSI) is presented by taking Channel Estimation (CE) errors as well as feedback

delay errors, deprived of considering the straight link from source to destination. An Energy Efficiency with Hybrid Fuzzy Firefly Algorithm (EE-HFFA) technique is provided in this research, for MIMO-AF based cooperative communication systems. By means of merging the benefits of the Firefly Algorithm (FA) as well as differential evolution (DE), EE-HFFA algorithm is presented. The outcomes of these two techniques are based upon the fuzzy membership function. Branch Convolutional Neural Network (B-CNN) classifier is presented for approximation of PCSI for the source node as well as relay nodes to maximize the capability of cooperative MIMO-AF systems.

3. Proposed methodology

A new MIMO-AF system under Partial Channel State Information (PCSI) is presented in this research by means of taking Channel Estimation (CE) errors as well as feedback delay errors, deprived of considering the straight link from source to destination. An Energy Efficiency with Hybrid Fuzzy Firefly Algorithm (EE-HFFA) technique is provided in this research, for MIMO-AF based cooperative communication systems [23] in which just PCSI estimation is existing at the relays because of the high speed mobility. By means of merging the benefits of the Firefly Algorithm (FA) as well as differential evolution (DE), EE-HFFA algorithm is presented. The outcomes of these two techniques are based upon the fuzzy membership function. Branch Convolutional Neural Network (B-CNN) classifier is presented for approximation of PCSI for the source node as well as relay nodes to maximize the capability of cooperative MIMO-AF systems. Partial CSI at the receiver produces deviations at the evaluation of the channel at the receiver that reduces the channel strength and to saturate at some levels of Signal to Noise Ratio (SNR).

3.1. Cooperative MIMO-AF EE framework

The proposed method concentrates on energy utilization of classic non regenerative cooperative MIMO systems with 3 nodes, i.e.,Sc_N, Re_N and De_N, equipped with nSc_N, nRe_N and nDe_N antennas, correspondingly. As in [7], the transmission of data had been carried out in 3 steps of equal duration. The cumulative achievable rate in two time slots has been estimated for the MIMO-AF system [17] are provided, in two distinct means [24], as I(y, s)

$$X_{\Sigma}(X,Y) = w\log_2\left|I_{n_{Sc_N}} + X^{\dagger}\psi(Y)X\right| \tag{1}$$

$$\text{where } \psi(Y) = \sigma_0^2 Z_0^{\dagger} Z_0 + \sigma_2^{-2} Z_0^{\dagger} Y^{\dagger} Z_2^{\dagger} \Omega(Y)^{-1} Z_2 G Z_1 \tag{2}$$

$$\Omega(\mathrm{Y}) = \mathrm{I}_{\mathrm{nDe_N}} + \frac{\sigma_1^2}{\sigma_2^2} Z_2 Y Y^{\dagger} Z_2^{\dagger} \tag{3}$$

Also, *bw* is called the channel bandwidth, $X \in \mathbb{C}^{nSc_N \times nSc_N}$ and $Y \in \mathbb{C}^{nRe_N \times nRe_N}$ are the Sc_N and Re_N precoding matrices, respectively. Also, $Z_0 \in \mathbb{C}^{nDe_N \times nSc_N}$, $Z_1 \in \mathbb{C}^{nRe_N \times nSc_N}$, and $Z_2 \in \mathbb{C}^{nDe_N \times nRe_N}$ model the Sc_N to De_N, Sc_N to Re_N, and Sc_N to Re_NMIMO channels, correspondingly. Likewise, σ_0^2, σ_1^2 and σ_2^2 are known as the variations of the Gaussian noise at the Sc_N - De_Nlink), Re_N and De_N (Sc_N - Re_N - De_N link), correspondingly [23]. As a006E alternative, employ precoding matrices that improve the upper bound on the ergodic capacity. This method was presented as well as verified in [25] for identifying precoding with partial CSI. It is as well used in every step of the presented method for identifying $X \in \mathbb{C}^{nSc_N \times nSc_N}$ and $Y \in \mathbb{C}^{nRe_N \times nRe_N}$, correspondingly.

Partial CSI using Branch Convolutional Neural Network (B-CNN) classifier

Presume that the receiver contains incomplete insights of the statistics of the channel, which produces deviations at the approximation of the channel at the receiver; as well the antenna elements of the transmitter are spatially correlated. Develop a Branch Convolutional Neural Network (B-CNN) classifier that matches the performances of a simple MIMO versus AF systems. B-CNN as it encompasses numerous branch networks alongside the foremost convolution workflow to perform estimates hierarchically. In Fig. 1, the structural design of B-CNN is depicted. The B-CNN model is enthused by the idea from [26] that every layer of a CNN encompasses the hierarchical nature of the features in the network. CNN property is utilized to unite it with the former knowledge of class hierarchical structure to implement the network to learn human understandable conceptions in diverse layers. Alongside B-CNN, as well presented a new training technique that is modified to the B-CNN models called Branch Training strategy (BT-strategy). While training a B-CNN model with BT-strategy, lesser level parameters are triggered as well as trained formerly compared to the higher level parameters. This notion is enthused by the vanishing gradient problem [27] as well as the step-by-step training method espoused by Simonyan as well as Zisserman to train a deep ConvNet in [28]. BT strategy could thwart the consequence of vanishing gradient problem to a degree as well as enhance the performance of a B-CNN model.

The bottommost network is an arbitrary ConvNet. There are numerous branch networks and each of them outputs a coarse prediction. The last loss function is a weighted sum of all coarse losses. B-CNN model utilizes existent CNN components as building blocks to build a network with internal output branches. In Fig. 1a, the network at the bottom is a conventional CNN. It is a random ConvNet containing multiple layers. In Fig. 1, the middle part depicts the output branch networks of a B-CNN. Each branch net yields likelihood on the equivalent level in the label tree. On the top of all the branches, completely connected layers as well as a soft max layer are utilized to yield the output in one-hot depiction. Branch nets encompass ConvNets as well as completely connected neural networks. On the other hand for easiness, in the experimentations, we utilize completely connected neural networks as the branch nets. While performing classification, a B-CNN model yields as prediction outcomes of $X \in \mathbb{C}^{nSc_N \times nSc_N}$ as well as $Y \in \mathbb{C}^{nRe_N \times nRe_N}$, correspondingly. While the channels is fed into B-CNN, the network would yield the equivalent likelihoods as the data flow through as well as every level's loss would pay to the final loss function base on the loss weights distribution.

Loss function

The loss function of B-CNN is known as a weighted sum of all coarse as well as identifies prediction outcomes of $X \in \mathbb{C}^{nSc_N \times nSc_N}$ as well as $Y \in \mathbb{C}^{nRe_N \times nRe_N}$, correspondingly. The loss function is provided as follows

$$L_i = \sum_{k=1}^{K} -A_k \log\left(\frac{e^{f_{y_i}^k}}{\sum_j e^{f_{y_j}^k}}\right) \tag{4}$$

Here ire presents the ith channel in the mini-batch. K is known as the amount of coarse levels in the label tree as well a A_k is called the loss weight of kth level paying to the loss function. The term $-\log\left(\frac{e^{f_{y_i}^k}}{\sum_j e^{f_{y_j}^k}}\right)$ is known as the cross entropy loss of the ith channel on the kth level in the prediction as well as utilize f_j to represent the jth user in the vector 'f' of class scores, outputted by the final layer of the model. The loss function takes the complete levels' loss to guarantee the structure prior could act as a role of internal guide to the whole model as well as make it simple to flow the gradients back to the shallow layers.

Loss weight

In (4), Value A_k is known as the loss weight of every branch network in B-CNN. This value describes how much impact a level creates to the final loss function. Even if just comparative value matters how the loss function works utilize a standard depiction in the experimentations that each A_k must be a value amid 0 and 1 (both encompassed) as well as their summation must be 1.

The EE-SE trade-off and a power consumption model

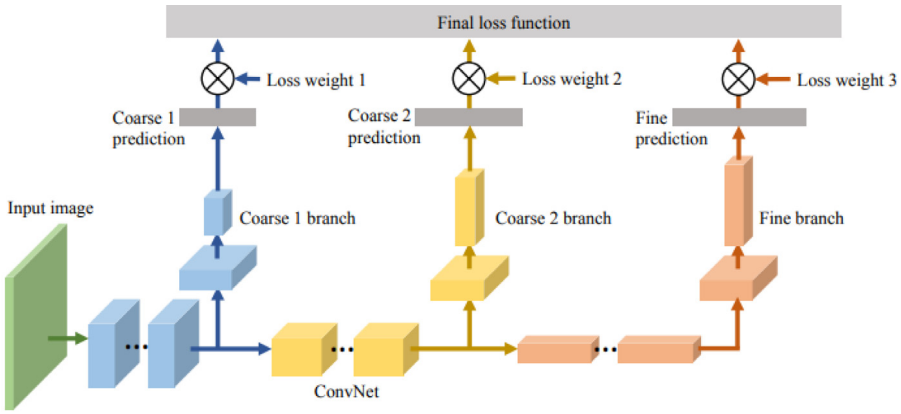


Fig. 1. Architecture of Branch Convolutional Neural Network (B-CNN).

Sc_N and De_N transmits in 2 phases such as means of power utilization utilizing every node can be distinct. In the primary stage, Sc_N transfers the data to Re_N and De_N, that get; merely Re_N transmits to De_N, thus Sc_N is idle while in the second phase. By considering the different power utilization modes i.e., the transmission, Po^{Tx}, the reception, Po^{Rx}, and the sleep, Po^{Sl}

It could henceforward be designed as

$$Po_{\Sigma} = \left(Po^{Tx}_{Sc_N} + Po^{Rx}_{Re_N} + Po^{Rx}_{De_N}\right) + \left(Po^{Sl}_{Sc_N} + Po^{Tx}_{Re_N} + Po^{Rx}_{De_N}\right) \tag{5}$$

A node generally takes baseband processing, RF transceiver chain and transferring of information. As per [24], the power utilization of general communication equipments/devices in cooperative MIMO-AF system, for instance BS, RN or the user equipment (UE), is linearly rely on the transmission power.

$$\mathrm{Po}^{\mathrm{Tx}} = \Delta\mathrm{Po} + \mathrm{nPo}^{\mathrm{CipA}} + \mathrm{Po}^{\mathrm{Ci}} \tag{6}$$

Here $\mathrm{Po}^{\mathrm{CipA}}$ designs the circuit power consumption of scaling with n. $\mathrm{Po}^{\mathrm{Ci}}$ designs other kinds of circuit power consumption. In case of reception mode, receiving node could takes power to obtaining (for instance RF transceiver chain) as well as processing the information (for example baseband processing), so

$$Po^{Rx} = \varsigma\left[nPo^{CipA} + Po^{Ci}\right] \tag{7}$$

Here $0 \le \varsigma \le 1$ provided where reception is typically in less demand regard to circuit power. Finally in the sleep mode, node would waits for transmit or receive and does not carry out any processing with the intension that just a fraction of circuit power is taken [29], that is to say $Po^{Sl} = nP^{SlpA}$, here P^{SlpA} is known as the per antenna sleep power. In implanting the definition of $\mathrm{Po}^{\mathrm{Tx}}$ in (4), $\mathrm{Po}^{\mathrm{Rx}}$ in (5) and $\mathrm{Po}^{\mathrm{Id}}$ in (3), over-all utilized power of cooperative MIMO-AF system has reformulated like

$$Po_{\Sigma} = Po_c + \Delta_{Sc_N} Po_{Sc_N} + \Delta_{Re_N} Po_{Re_N} \tag{8}$$

here, as in [7],

$$Po_{\mathrm{Sc_N}}(X) = tr\left\{XX^{\dagger}\right\} \tag{9}$$

$$Po_{\mathrm{Re_N}}(X,Y) = tr\left\{Y\left(\sigma_1^2 I_{n_{Re_N}} + Z_1 XX^{\dagger} Z_1^{\dagger}\right)Y^{\dagger}\right\} \tag{10}$$

Furthermore,

$$Po_{\mathrm{c}} = n_{Sc_N}\left(Po^{CipA}_{Sc_N} + Po^{SlpA}_{Sc_N}\right) + Po^{Ci}_{Sc_N} + (1+\varsigma)\left(n_{Re_N} Po^{CipA}_{Re_N} + Po^{SlpA}_{Re_N}\right) + 2\varsigma\left(n_{De_N} Po^{CipA}_{De_N} + Po^{SlpA}_{De_N}\right) \tag{11}$$

represent the entire fixed circuit consumed powers. Henceforth, dependent upon (6) and (7), Po_{Σ} is stated in two diverse ways as

$$Po_{\Sigma}(X,Y) = Po'_c(X) + \frac{\sigma_1^2}{\sigma_2^2} + Tr\left(Y\ddot{\gamma} + Po^{SlpA}_{De_N}\right) \tag{12}$$

$$Po'_c(X) = Po_c + \Delta_{Sc_N} Po_{Sc_N}(X) \tag{13}$$

$$Po'_c(X) = Po_c + \Delta_{Re_N}\sigma_1^2 tr\left\{XX^{\dagger}\right\} \tag{14}$$

$$\ddot{\gamma}(X) = \frac{\sigma_1^2}{\sigma_2^2}\Delta_{Re_N}\left(\sigma_1^2 I_{n_{Re_N}} + H_1 XX^{\dagger} Z_1^{\dagger}\right) \tag{15}$$

$$\ddot{\psi}(\alpha,\beta,X) = \alpha I_{n_{Sc_N}} + \beta Z_1^{\dagger} Y^{\dagger} Y Z_1 \tag{16}$$

The energy utilization, $\mathrm{EE_b}$, (or EE, $1/\mathrm{EE_b}$) as a ratio among consumed power as well as rate, the EE-SE tradeoff of cooperative MIMO-AF system is defined on this way

$$EE_b(X,Y) = \frac{Po_{\Sigma}(X,Y)}{X_{\Sigma}(X,Y)} \tag{17}$$

Here, in (1) and (7), thorough expressions for $Po_{\Sigma}(X,Y)$ and $X_{\Sigma}(X,Y)$ are provided.

c. EE-optimal source and relay precoding

The main intention is to determine X as well as Y that are precoding matrices and decreases the consequent optimization problem

$$\min_{X,Y} EE_b(X,Y) \tag{18}$$

$$s.t.\ Po_{Sc_N}(X) \le Po^{max}_{Sc_N} \tag{19}$$

$$Po_{Re_N}(X) \le Po^{max}_{Re_N} \tag{20}$$

Here $X \ge 0, X = 0$, and Y = 0. Moreover, $Po^{max}_{Sc_N}$ and $Po^{max}_{Re_N}$ are called the greatest transmit power at Sc_N and Re_N. As Compared to sum-rate maximization [30], or transmit power and MSE minimization problems [8], EE objective function in (18) is not schur-concave/convex objective function [17], but a proportion amongst the Schur-concave/convex functions. Basically, contrary to Schur concave/convex in objective functions, (20) displays global optimum [31] (that is not 0 or 1 as long as $\mathrm{Po_c} > 0$) even when no constraints are enforced.

d. EE-optimal source precoding

In fact, in keeping with [5], can be solved in its unrestricted form [5], specifically solving (18) equations of (19) and (20). Primarily, derive X^* for a known Y and Y^* for a known X. By using the alterations of variables $H = \ddot{\psi}^{\frac{1}{2}} XX^{\dagger}\ddot{\psi}^{\frac{1}{2}}$ $(H \ge 0, H \ne 0)$ is called a hermitian

matrix and $\psi(\alpha,\beta) = \left(\dot{\psi}^{-1}\right)^{\frac{1}{2}} \ddot{\psi}(\alpha,\beta) \left(\dot{\psi}^{-1}\right)^{\frac{1}{2}}$ the optimization problem is specified in this manner,

$$H^* = \arg\min_{H} EE_b(H) = \frac{Po'_c + \Delta_{Sc_N} Po_{Sc_N}(H) + \Delta_{Re_N} Po'_{\mathrm{Re_N}}(H)}{X_{\Sigma}(H)} \tag{21}$$

$$\text{s.t } Po_{Sc_N} \le Po^{max}_{Sc_N} \& Po'_{\mathrm{Re_N}}(H) \le Po^{max}_{\mathrm{Re_N}}(H) \tag{22}$$

$$\text{where } X_{\Sigma}(H) = W \log_2 \left| I_{n_{Sc_N}} + H \right| \tag{23}$$

$$Po_{\mathrm{Sc_N}}(H) = tr\{H\psi(1,0)\} \tag{24}$$

$$Po'_{\mathrm{Re_N}}(H) = tr\{H\psi(0,1)\} \tag{25}$$

e. EE-optimal relay precoding

Based upon Eqs. (17)–(18), (12) and (1), the precoding matrix Y^* is calculated as

$$Y^* = \underset{Y}{\operatorname{argmin}}\, EE_b(X,Y) = \frac{Po'_C(X) + \frac{\sigma_1^2}{\sigma_2^2}\left\{Y\ddot{\gamma}(X)Y^{\dagger}\right\}}{X_0(X) + W log_2 |\Omega(Y|\gamma(X)^{1/2})\Omega(G)^{-1}} \tag{26}$$

Here

$$\gamma(X) = I_{n_{Re_N}} + \dot{\gamma}(X) \tag{27}$$

As per the aforementioned equations, the EE-optimal relay precoding matrix, Y^*, are obtained with the help of Energy Efficiency with Hybrid Fuzzy Firefly Algorithm (EE-HFFA) technique beside like gradient search unrestricted case or in expected gradient search by means of power restrained case method for achieving $G(EE_b^{lr} + \varepsilon^*)$ for a given value of ε^*. The search for optimal ε^* ends while difference amongst EE_b and $EE_b^{lr} + \varepsilon^*$ become irrelevant are conjointly satisfied.

Energy Efficiency with Hybrid Fuzzy Firefly Algorithm (EE-HFFA) scheme

Firefly Algorithm (FA) [32] is a novel biologically enthused metaheuristic optimization technique enthused by the flashing conduct of tropical fireflies. Differential evolution (DE) [33] is as well a metaheuristic algorithm. DE with a possible parallel structure is a non-gradient- based, evolutionary computation technique. FA is carried out in keeping with the communication behavior of tropical fireflies as well as the idealized conduct of the flashing patterns. In this proposed methodology, the Firefly is considered as the node position for finding out the best possible EE value in Eqs. (18), (21) and (26). FA utilizes the subsequent three idealized rules [34] for identifying the best possible value of EE with subsequent steps:

- Each and every fireflies are unisex with the intension that one firefly would be fascinated to other fireflies irrespective of their sex;
- Attractiveness is relative to their brightness. Therefore the less bright one would move in the direction of the brighter one for any two flashing fireflies. The attractiveness is relative to the brightness and they both reduce as their distance rises;
- The brightness of a firefly is distressed or identified by the landscape of the EE value.

Two significant points are there in the standard FA. First is the formulation of the light intensity and next is the modification of the attractiveness. Initially, we could constantly presume that the brightness of the firefly is identified by the encoded EE value in Eqs. (18), (21) and (26). Next, have to describe the change of light intensity as well as formulate the modification of the attractiveness. In nature the light intensity reduces with the distance from its source as well as the media would engross the light intensity I changes with the distance r as well as light absorption parameter γ exponentially as well as monotonically [35].

$$I = I_0 e^{-\gamma r^2} \tag{28}$$

here I_0 is known as the original light intensity at the source is identified via the Eqs. (18) (21) (26) and γ is called the light absorption coefficient. As a result describe the firefly's light attractive coefficient β in the identical manner as the light intensity coefficient I.

$$\beta = \beta_0 e^{-\gamma r^2} \tag{29}$$

here β_0 is called the original light attractiveness at $r = 0$. In order to compute the distance amid any two node positions I and j at x_i and x_j, the Cartesian distance is utilized.

$$r_{ij} = \left| |x_i - x_j| \right|_2 \tag{30}$$

where d is the number of nodes. The amount of movement of node $i$to another more attractive (brighter) node j is determined by

$$x_i = x_i + \beta_0 e^{-\gamma r^2}\left(x_j - x_i\right) + \alpha\varepsilon_i \tag{31}$$

here the x_i first term is known as the present location of node i, the second term β_0 is because of the attraction, when the third term α is known as randomization with the vector of random variables ε_i being drawn from diverse distributions for instance the Uniform distribution, Gaussian distribution and Lévy flight. In the third term, α is known as a scaling parameter, which regulates the step size as well as it must be associated with the interests of the problems.

Differential evolution

Differential evolution (DE) utilizes a vectorized mutation operator as well as different forms of crossover to progress from the arbitrarily produced, preliminary starting points to the probably best possible solution. In this research method, DE/rand/1/bin scheme/variant is utilized. This variant is extensively utilized and it is shortly explained in this manner [36]. A population encompasses n number of nodes for a provided D-dimensional minimization problem. The mutant vector mv_i is described in this manner

$$mv_{i,g+1} = x_{r_1,g} + F\left(x_{r_2,g} - x_{r_3,g}\right) \tag{32}$$

Here the indexes r_1, r_2, $r_3 \in [1, n]$ relate to three EE value solutions arbitrarily selected from the complete nodes position as well as g is known as the generation index. Furthermore, F (F ∈ [0,2]) is known as a perturbation parameter, which regulates the amplification of the difference vector $x_{r_2,g} - x_{r_3,g}$, however in maximum cases $0 < F < 1$ is utilized. The binomial crossover operation focuses on generating a novel trial vector from the perturbed or mutated vector $mv_{i,g+1}$ and the target vector $x_{i,g}$

$$u_{i,g+1} = \begin{cases} mv_{ij,g+1}, if\ r(j) \le C_r\ or\ j = random(i) \\ mv_{ij,g}, if\ r(j) > C_r\ or\ j \ne random(i) \end{cases} \tag{33}$$

Here $j \in [1,2,\ldots,D], r(j)$ is known as the jth realization of a uniform random generator number. Furthermore, $C_r \in [0,1]$ is known as crossover constant. $random \in [1,2,\ldots,D]$ is known as a random permutation index vector that could guarantee that the trial vector ui, $g + 1$ obtains as a minimum of one character from the mutated vector $mv_{i,g+1}$. The selection technique is alike to those of other techniques where a greedy acceptance is carried out:

$$x_{i,g+1} = \begin{cases} u_{i,g+1}, if\ f\left(u_{i,g+1}\right) \le f\left(x_{i,g}\right) \\ x_{i,g}, elsewhere \end{cases} \tag{34}$$

This signifies that the update is agreed only when an improved objective is attained. Hybrid Fuzzy Firefly Algorithm (EE-HFFA) dependent upon FA and DE as to raise the convergence speed as well as the diversity of the nodes population. Amongst numerous components of techniques, intensification and diversification are considered as the two

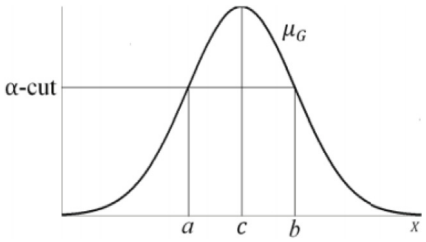


Fig. 2. Gaussian membership function.

basic elements of any meta-heuristic algorithm [37]. With the intention of exploring the search space globally, meta-heuristic algorithms must produce varying number of solutions with the help of diversification or exploration technique. The advantages of the proposed system are that FA is good at exploration as well as diversification. Furthermore, technically speaking, due to the efficiency of mutation operator and crossover operator, differential evolution can provide a good mixing ability among the population and thus provide a better diversity in the population

Proposed algorithm

Initially, the former literature denoted that the firefly algorithm could categorize the complete nodes into smaller groups robotically in regard to the attraction technique through the light intensity deviation as well as one among the FA variants could escape from the local minima because of long-distance mobility by Lévy flight [38] that resolves exploration and diversification problems in FA. Also, DE could give a better mixing capability amongst the population because of the efficacy of mutation operator as well as crossover operator, and therefore offer an improved diversity in the population. Furthermore, bringing up to date, the current global best in the whole EE best possible value guarantees that solutions could converge to the optimum, when diversification by means of mixing as well as regrouping the entire channels or nodes lets the search algorithm to escape from local optima and might concurrently raise the diversity of solutions. The foremost dominance of such mixing as well as regrouping is to assure the search concentrating on the present node position in the hopeful improved EE optimal value.

The outcomes of the FA and DE are united by utilizing the fuzzy membership function.

Gaussian fuzzy sets

The Gaussian membership function [39,40] is as follows

$$EE_{new} = \mu_G(EE, c, \sigma) = e^{-[EE-c]^2/2\sigma^2} \tag{35}$$

Here c is known as the center of the fuzzy set, and σ parameterizes the width of the fuzzy set. The α cut a fuzzy set is a crisp set, which encompasses the entire elements of the universe set X that contain a membership grade equivalent to or superior to α. Dependent upon the α cut of a fuzzy set, the width of the fuzzy set, W_α is identified.

According to Fig. 2, W_α is ac or cb that is identified by the subsequent equation (36)

$$W_\alpha = ac = cb = \sqrt[2]{(-\ln\alpha)\left(2\sigma^2\right)} \tag{36}$$

According to the above mentioned steps, the simulation outcomes of modest MIMO and MIMO-AF systems saturate while the receiver of the system contains just partial statistical knowledge of the channel.

4. Results and discussions

The simulations consider the downlink transmission of cooperative MIMO AF system. Thus Sc_N is known as a BS and De_N is called as UE. Also based on the attributes in Table 1 [41] but having $P_{Sc_N}^{SlA} = 0$ W to set values for entire parameters. Also Po^{max} for all antennas is amongst

Table 1
Characteristics of base stations installed on analyzed site.

Characteristics of on-site base stations	Base station model (UMTS)
Year of base station model production	2010
Frequency band	2100 MHz
Number of base station racks	1
Number of transceivers in one base station rack	3
Number of sectors covered by one rack	3
Number of transceivers per sector	1/1/1
Number of antennas per sector	1
Number of combiners in rack	Integrated in RF module
Number of antenna cables per antenna	2
Output power per sector	25 W
Antenna cable diameter	7/8″
Average power consumption of site	7.5 kW

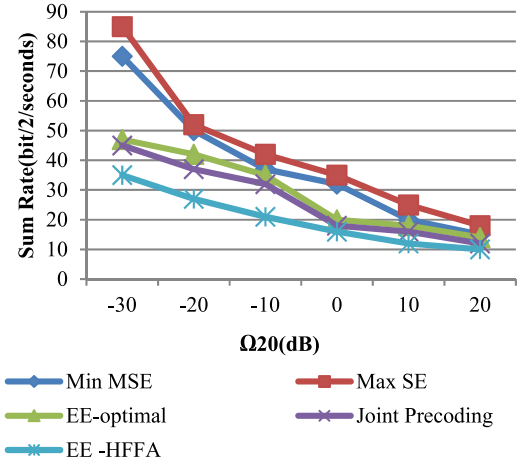


Fig. 3. Sum rate vs. EE-HFFA schemes (σ_0^2 dB).

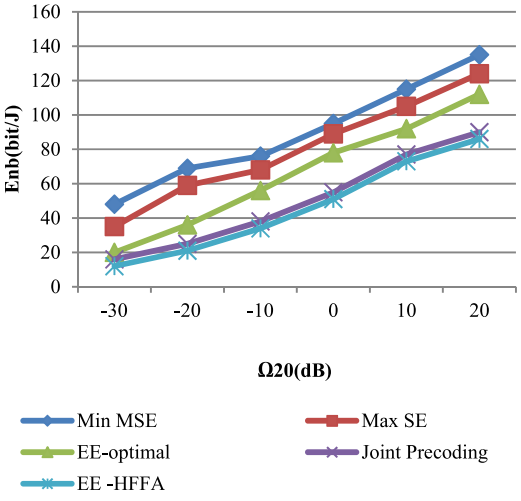


Fig. 4. Energy vs. EE-HFFA schemes (σ_0^2 dB).

6 to 20 W for typical micro/ macro BS [2], and amongst 1 to 5 W for typical urban/ rural RN [24]. Let consider single-tap i.i.d MIMO Rayleigh fading channel amongst every node, a unit bandwidth such as b = 1. Moreover, we assume that the complete channel matrices encompass a like dimension, i.e. n = nSc_N = nRe_N = nDe_N, and N = 10 in step 1).

Figs. 3 and 4 compares different variables in the EE-HFFA technique, which does not conform to the current precoding methods that are allotted in the sense of decrease the MSE [24] or rise SE [42] of the cooperative MIMO-AF systems for n = 4, $P_{Sc_N}^{max}$ = 80 W, $P_{Re_N}^{max}$ = 10

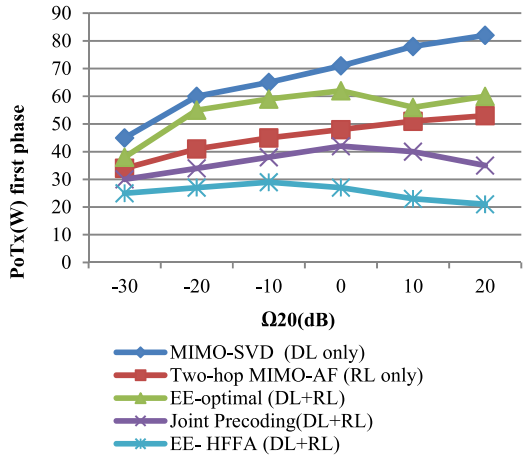


Fig. 5. Transmit power (first phase) vs. EE-HFFA based DL only and RL only schemes.

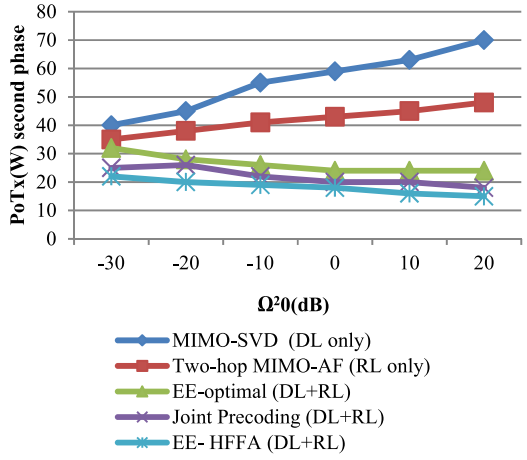


Fig. 6. Transmit power (second phase) vs. EE-HFFA based DL only and RL only schemes.

W, $\sigma_0^2 = \sigma_1^2 = \sigma_2^2/10$. It is anticipated, the precoding methods of [24] and [5] provide as well as the presented EE-HFFA method carries out healthier while matched up with MSE as well as sum-rate performances. In Fig. 4 energy consumption results of the proposed EE-HEFA system consumes lesser energy of 86 J whereas other Min MSE, Max SE, EE-optimal, Joint precoding consumes higher energy values of 135 J, 124 J, 112 J and 90 J respectively.

Figs. 5–8 shows the evaluation of the transmission power and the performance of EE-HFFA method, compared to other methods [43] and RL [42] by own, cognitive MIMO AF relaying systems [44] for n = 4, $P_{Sc_N}^{max} = 80$ W, $P_{Re_N}^{max} = 10$ W, $\sigma_0^2 = \sigma_1^2 = \sigma_2^2/10$. In Fig. 5 transmit power (first phase) results of the proposed EE-HEFA system consumes lesser value 15 W, whereas other MIMO-SVD, Two-hop MIMO-AF, EE-optimal, Joint precoding consumes higher power values of 70 W, 48 W, 24 W and 18 W respectively.

In Fig. 6 transmit power (first phase) results of the proposed EE-HEFA system consumes lesser value 21 W, whereas other MIMO-SVD, Two-hop MIMO-AF, EE-optimal, Joint precoding consumes higher power values of 82 W, 53 W, 60 W and 35 W respectively.

In Fig. 7 sum rate results of the proposed EE-HEFA system consumes lesser value 12 bit/s, whereas other MIMO-SVD, Two-hop MIMO-AF, EE-optimal, Joint precoding needs higher sum rate values of 25 bit/s, 20 bit/s, 16 bit/s and 15 bit/s respectively.

In Fig. 8 energy comparison results of the proposed EE-HEFA system consumes lesser value 120 J W, whereas other MIMO-SVD, Two-hop

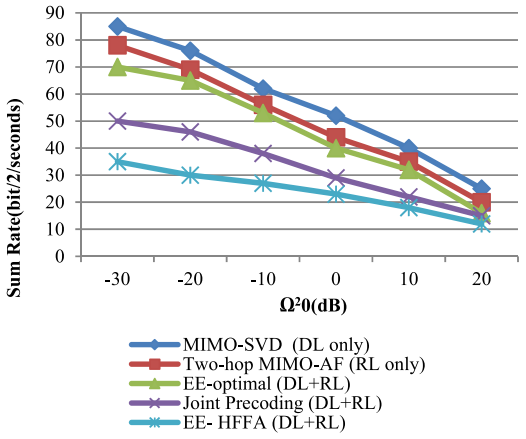


Fig. 7. Sum rate vs. EE-HFFA based DL only and RL only schemes.

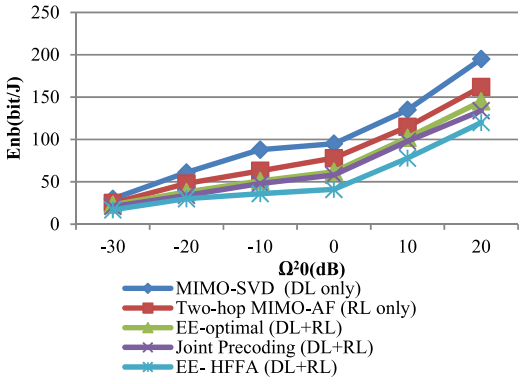


Fig. 8. Energy vs. EE-HFFA that depends on the DL only and the RL only approaches.

MIMO-AF, EE-optimal, Joint precoding consumes higher power values of 195 J, 162 J, 145 J and 134 J respectively. Power restrained results are intended for all methods. As per En_b performance, the results show that new EE-HFFA method outperforms when matched up with the other methods. As per the sum-rate, in this particular setting, the RL + DL with EE-HFFA method does superior when matched up with other methods. Definitely, the transmit power results state that the DL transmission at all times plentiful transmit power when compared to other three methods, successively, this defines the cause for less energy efficiency matched up with EE-HFFA method.

5. Conclusion and future work

The proposed method (that is to say while relay as well as direct links are taken) was presented for Partial Channel State Information (PSCI) dependent upon EE optimal source as well as relay precoding matrices. In this work, UAVs as amplify-and-forward MIMO relays toward give connectivity among an obstructed multi-antenna equipped source and destination. The consequence of PSCI at the receiver is estimated by means of the application of Branch Convolutional Neural Network (B-CNN) classifier that produces estimation error of the channel at the receiver as well as the spatial correlation on MIMO-AF scenario systems. This proposed method was enthused by a natural insight that there is a tradeoff amid the development in channel capacity as well as the amount of broadness of the PCSI at the transmitter. Partial CSI at the receiver produces errors at the approximation of the channel at the receiver that yields the channel capacity to decrease and to saturate at some levels of SNR. Furthermore, properly shown the optimality of source as well as relay precoders, while handled self-reliantly, with the help of pseudo-convexity/convexity arguments as

well as have based upon Hybrid Fuzzy Firefly Algorithm Based Energy Efficiency (EE-HFFA) optimization for jointly enhancing them. The given closed-form expression for the EE-optimal source precoders as well as have developed an optimal numerical method in the sense of getting relay pre coder. Prolonging the present work for case in which just the imperfect or the partial CSI that exists in source or relay node for Interference Reduction or noise reduction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

S. Kanithan: Conceptualization, Methodology, Software, Writing - original draft, Data curation. **N. Arun Vignesh:** Writing - review & editing. **E. Karthikeyan:** Supervision, Validation. **N. Kumareshan:** Investigation.

References

[1] H. Zhang, A. Gladisch, M. Pickavet, Z. Tao, W. Mohr, Energy efficiency in communications, IEEE Commun. Mag. 48 (11) (2010) 48–49.

[2] G. Auer, V. Giannini, C. Desset, I. Godor, P. Skillermark, M. Olsson, M.A. Imran, D. Sabella, M.J. Gonzalez, O. Blume, A. Fehske, How much energy is needed to run a wireless network?, IEEE Wirel. Commun. 18 (5) (2011) 40–49.

[3] J.N. Laneman, D.N. Tse, G.W. Wornell, Cooperative diversity in wireless networks: Efficient protocols and outage behavior, IEEE Trans. Inf. Theory 50 (12) (2004) 3062–3080.

[4] Reconfigurable OFDMA-Based Cooperative NetworKs Enabled By Agile SpecTrum Use (ROCKET), ICT-215282 FP7 STREP project, Tech. Rep., 2008-2009, [Online]. Available: http://www.ict-rocket.eu.

[5] F. Héliot, Low-complexity energy-efficient joint resource allocation for two-hop MIMO-AF systems, IEEE Trans. Wirel. Commun. 13 (6) (2014) 3088–3099.

[6] 3GPP TS 36.216 (2010-15), Evolved Universal Terrestrial Radio Access (E-UTRA); Physical Layer for Relaying Operation (Release 10-13), 3GPP, Tech. Rep., 2010-2015, [Online]. Available: http://www.3gpp.org/dynareport/36216.htm.

[7] O. Muñoz Medina, J. Vidal, A. Agustin, Linear transceiver design in nonregenerative relays with channel state information, IEEE Trans. Signal Process. 55 (6) (2007) 2593–2604.

[8] Y. Rong, Optimal joint source and relay beamforming for MIMO relays with direct link, IEEE Commun. Lett. 14 (5) (2010) 390–392.

[9] C. Xing, S. Ma, Z. Fei, Y.-C. Wu, H.V. Poor, A general robust linear transceiver design for multi-hop amplify-and-forward MIMO relaying systems, IEEE Trans. Signal Process. 61 (5) (2013) 1196–1209.

[10] A. Zappone, P. Cao, E.A. Jorswieck, Energy efficiency optimization in relay-assisted MIMO systems with perfect and statistical CSI, IEEE Trans. Signal Process. 62 (2) (2014) 443–457.

[11] Y. Hua, An overview of beamforming and power allocation for MIMO relays, in: Proceedings of the IEEE Military Communications Conference, 2010, pp. 375–380.

[12] K. Lee, S.-H. Park, J.-S. Kim, I. Lee, Degrees of freedom on MIMO multi-link two-way relay channels, in: Proceedings of the 53rd IEEE Global Communications Conference, 2010, pp. 1–5.

[13] J. Joung, A.H. Sayed, Multiuser two-way amplify-and-forward relay processing and power control methods for beamforming systems, IEEE Trans. Signal Process. 58 (3) (2010) 1833–1846.

[14] B.K. Chalise, L. Vandendorpe, Y.D. Zhang, M.G. Amin, Local CSI based selection beamforming for amplify-and-forward MIMO relay networks, IEEE Trans. Signal Process. 60 (5) (2012) 2433–2446.

[15] H.W. Je, D.H. Kim, K.B. Lee, Joint precoding for MIMO-relay systems with partial channel state information. in: IEEE International Conference on Communications, 2009(ICC'09), 2009, pp. 1-5.

[16] T. Kanakis, P.B. Rapajic, MIMO relay channels with partial channel knowledge/estimation error and spatial correlation, in: International Conference on Signal Processing and Communication Systems.

[17] F. Héliot, Tafazolli, Optimal energy-efficient source and relay precoder design for cooperative MIMO-AF systems, IEEE Trans. Signal Process. 66 (3) (2017) 573–588.

[18] Lina Zhang, LiqiangLiu, Xin-She Yang, Yuntao Dai, A novel hybrid firefly algorithm for global optimization, PLoS One (2016) http://dx.doi.org/10.1371/journal.pone.0163230.

[19] S. Kanithan, G.M. Tamilselvan, A novel enhanced bat optimization based energy efficiency and imperfect channel state information in cooperative MIMO-AF systems, Concurr. Comput.: Pract. Exper. (2018).

[20] TriantafyllosKanakis, Predrag B. Rapajic, Rapajic MIMO relay channels with partial channel knowledge/estimation error and spatial correlation, 2007.

[21] Sinchu Padmanabhan, Reuben Stephen, Chandra Murthy, Marceau Coupechoux, Training-based antenna selection for per minimization: A POMDP approach, IEEE Trans. Commun. 63 (2015) 3247–3260, http://dx.doi.org/10.1109/TCOMM.2015.2455504.

[22] Shihao Yan, Nan Yang, Robert Malaney, Jinhong Yuan, Transmit antenna selection with alamouti coding and power allocation in MIMO wiretap channels, IEEE Trans. Wireless Commun. 13 (2014) 1656–1667, http://dx.doi.org/10.1109/TWC.2013.013014.131248.

[23] Z. He, W. Jiang, Y. Rong, Robust design for amplify-and-forward MIMO relay systems with direct link and imperfect channel information, IEEE Trans. Wireless Commun. 14 (1) (2015) 353–363.

[24] Y. Qi, F. Héliot, M.A. Imran, Green relay techniques in cellular systems, in: F.R. Yu, X. Zhang, V.C. Leung (Eds.), Green Communications and Networking, CRC press, Boca Raton, FL, 2012, ch. 3.

[25] H.W. Je, B.O. Lee, K.B. Lee, Relay weighting for non- regenerative MIMO-relay system with partial channel state information, IEEE Trans. Wire. Commun. (2009) submitted for publication.

[26] M.D. Zeiler, R. Fergus, Visualizing and Understanding Convolutional Networks, Springer International Publishing, Cham, 2014, pp. 818–833.

[27] S. Hochreiter, Y. Bengio, P. Frasconi, J. Schmidhuber, Gradient flow in recurrent nets: the difficulty of learning long-term dependencies, 2001.

[28] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, CoRR, abs/1409.1556.

[29] G. Miao, N. Himayat, G.Y. Li, Energy-efficient link adaptation in frequency-selective channels, IEEE Trans. Commun. 58 (2) (2010) 545–554.

[30] F. Héliot, R. Hoshyar, R. Tafazolli, On the spectral efficiency maximization of nonregenerative cooperative MIMO communication systems, in: Proc. IEEE PIMRC, Istanbul, Turkey, 2010.

[31] F. Héliot, M.A. Imran, R. Tafazolli, On the energy efficiency spectral efficiency trade-off over the MIMO Rayleigh fading channel, IEEE Trans. Commun. 60 (5) (2012) 1345–1356.

[32] X.S. Yang, Firefly Algorithm, in: Nature-Inspired Metaheuristic Algorithms, Luniver Press, 2010, pp. 79–90.

[33] R. Storn, K. Price, Differential evolution a simple and efficient heuristic for global optimization over continuous spaces, J. Glob. Optim. 11 (4) (1997) 341–359.

[34] H. Wang, Z. Cui, H. Sun, S. Rahnamayan, X. Yang, Randomly attracted firefly algorithm with neighborhood search and dynamic parameter adjustment mechanism, Soft Comput. (2016) 1–15.

[35] X.S. Yang, Firefly algorithm, in: Leâvy Flights and Global Optimization, in: Research and Development in Intelligent Systems, vol. XXVI, Springer London, 2010, pp. 209–218.

[36] K. Price, R.M. Storn, J.A. Lampinen, Differential Evolution: A Practical Approach to Global Optimization, Springer Science & Business Media, 2006.

[37] C. Blum, A. Roli, Metaheuristics in combinatorial optimization: Overview and conceptual comparison, ACM Comput. Surv. (CSUR) 35 (3) (2003) 268–308.

[38] X.S. Yang, Cuckoo Search and Firefly Algorithm: Overview and Analysis, Cuckoo Search and Firefly Algorithm, Springer International Publishing, 2014, pp. 1–26.

[39] C.M. Lin, H.Y. Li, Adaptive dynamic sliding-mode fuzzy CMAC for voice coil motor using asymmetric Gaussian membership function, IEEE Trans. Ind. Electron. 61 (10) (2014) 5662–5671.

[40] K.H. Cheng, C.F. Hsu, C.M. Lin, T.T. Lee, C. Li, Fuzzy–neural sliding-mode control for DC–DC converters using asymmetric Gaussian membership functions, IEEE Trans. Ind. Electron. 54 (3) (2007) 1528–1536.

[41] J. Lorincz, T. Garma, G. Petrovic, Measurements and modelling of base station power consumption under real traffic loads, Sensors 12 (4) (2012) 4281–4310.

[42] R. Mo, Y.H. Chew, Precoder design for non-regenerative MIMO relay systems, IEEE Trans. Wirel. Commun. 8 (10) (2009) 5041–5049.

[43] R.S. Prabhu, B. Daneshrad, Energy-efficient power loading for a MIMO-SVD system and its performance in flat fading, in: Proc. IEEE Globecom, Miami, USA, 2010.

[44] X. Jia, M. Xie, L. Yang, H. Zhu, Outage performance and ergodic capacity of multiuser cognitive MIMO AF relaying systems with MUS/mrt/RAS over nakagami-m fading channels, Trans. Emerg. Telecommun. Technol. 28 (3) (2017) 1–23.